

SYSTEMS FOR APPLIED MACHINE LEARNING

Technical Report Analysis

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Systems for Applied Machine Learning:

Designing, Build, and Critique of NLP
Sentiment Analysis with Spam Handling and
Computer Vision Face Alignment under
Image Distortion

Machine Learning Report

Module: Applied Machine Learning
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AI Usage Statement

I acknowledge the use of Microsoft copilot and Perplexity as an AI assistant during the development of this assignment. The AI was used specifically for the following purposes:

- Providing guidance on how to structure code comments clearly and professionally.
- Suggesting a logical and sequential structure for the notebook/code.
- Offering debugging support and troubleshooting advice when encountering errors related to model training, accuracy, and evaluation metrics.
- Helping with the formatting and presentation of results and figure captions.

All core implementation, experimental design, data analysis, model development, and final results were carried out by me. The AI was used as a supportive tool similar to a tutor or coding assistant, and I take full responsibility for the work submitted.

Date: 22 May 2026

1 INTRODUCTION

This report presents the design, implementation, and evaluation of two applied machine learning systems in **Natural Language Processing (NLP)** and **Computer Vision (CV)**. The first investigates sentiment classification of movie reviews under deliberate dataset contamination, where email-type spam is randomly distributed across sentiment classes, while the second examines facial-landmark alignment under geometric variation and image distortion, including rotations, brightness shifts, and noise. The overall aim is to build well-justified pipelines that maintain reliable predictive performance under noisy and variable input conditions.

These tasks reflect practical machine learning challenges. Sentiment analysis supports applications such as user-feedback monitoring and recommendation refinement, while face alignment is a core component of facial recognition, augmented reality, and medical imaging pipelines.

In line with the assignment brief, all models were trained and evaluated only on the supplied datasets. For Task 1, classical machine learning methods were chosen over transformer-based models to prioritise interpretability, computational economy, and controlled model comparison, while Task 2 focuses specifically on facial-landmark localisation, excluding face detection to isolate alignment performance.

The remainder of the report is organised as follows. **Section 2** presents the dataset analysis, preprocessing, methodology, and evaluation for the sentiment analysis system, while **Section 3** describes the methods, results, and robustness analysis for the face alignment system.

2 PART 1: SENTIMENT ANALYSIS IN MOVIE REVIEWS

2.1 METHODOLOGY

Exploratory Data Analysis and Spam Identification

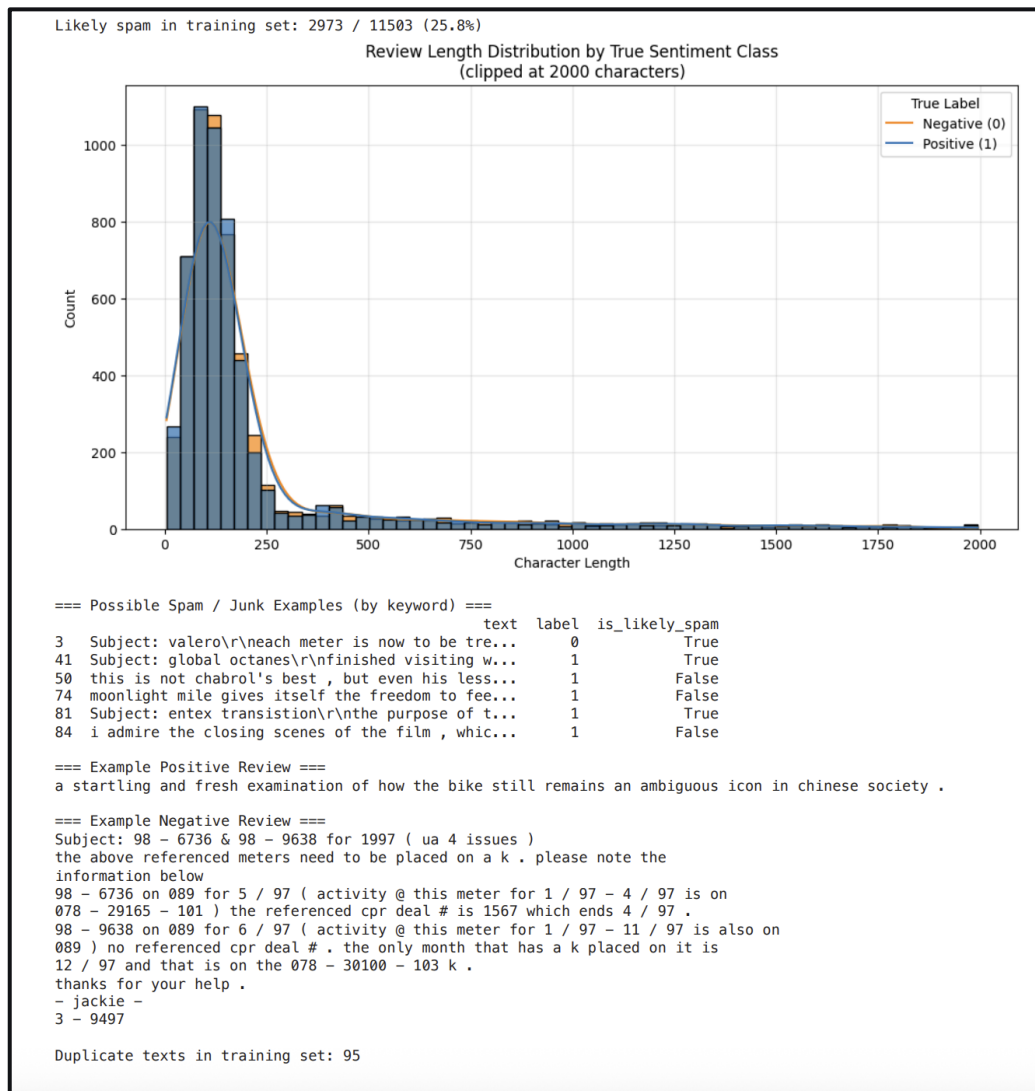
Task 1 concerns the development of a binary sentiment classification system for predicting whether a movie review is positive (label 1) or negative (label 0). A key challenge in the dataset is the deliberate random injection of email-style spam within both sentiment classes. Because such spam does not convey authentic opinions about films, a proportion of the assigned labels are effectively unreliable or semantically meaningless. This introduces substantial label noise, thereby increasing the risk that conventional supervised learning methods will capture misleading correlations associated with promotional or administrative language rather than genuine sentiment cues.

The dataset comprises 11,503 training samples, 1,397 validation samples, and 1,434 test samples. The class distributions in both the training and validation partitions are approximately balanced, with close to equal proportions (~50%) of positive and negative examples. Exploratory data analysis indicated that around 25.8% of the training instances (2,973 samples) exhibit clear spam-like characteristics, primarily identified through email-specific formatting signals such as the “Subject:” prefix and repeated line-break patterns (`\r\n`). In addition, duplicate analysis identified 95 repeated texts, further suggesting the presence of recycled or non-review material within the corpus.

As illustrated in Figure 1, the distribution of review lengths by sentiment label reveals a distinct cluster of unusually short texts that overlaps strongly with the subset flagged as likely spam. This pattern provides further evidence that the dataset is contaminated by out-of-domain content rather than consisting exclusively of authentic movie reviews.

Taken together, these observations motivate the use of a spam-aware modelling strategy rather than a standard off-the-shelf sentiment classifier. The approach adopted in this work therefore combines conventional sentiment classification with explicit spam detection and dummy labelling to reduce the influence of contaminated, non-review samples on the learning process.

Figure 2.1.1. Distribution of review lengths by true sentiment label in the training set (clipped at 2000 characters). *Short texts form a distinct cluster highly associated with spam samples.*



Text Pre-processing Strategy and Justification

Prior to feature extraction, all text samples were processed with a custom `preprocess_text` function designed to reduce structural noise while preserving linguistically informative content in the contaminated dataset. Email-style formatting artefacts, including `\r\n` and `\n\n` line breaks, were normalised to single spaces to improve textual consistency, and all text was converted to lowercase to ensure that orthographic variants of the same word map to a single feature. URLs, email addresses, and numeric sequences were replaced with the placeholder tokens `url`, `email`, and `num` respectively, preventing highly specific low-frequency strings from inflating the feature space while preserving structural cues useful for identifying spam-like content. Non-alphabetic characters were removed except for (`!`, `?`, `.`, and `,`), retained as markers of sentiment intensity, and excess whitespace was stripped.

This conservative normalisation strategy was motivated by the dataset's contamination profile and by standard NLP practice, in which extensive preprocessing is necessary to reduce noise while preserving semantically relevant information [8] [17]. Stemming and lemmatisation were deliberately excluded: while they can reduce vocabulary size, collapsing inflected word forms risks conflating terms with distinct sentiment polarity and offers limited benefit when short n-gram phrases are the primary feature unit [8]. Preserving original word forms was therefore more appropriate for the TF-IDF n-gram configuration described in the following section.

Feature representation

The cleaned text was represented using TF-IDF vectorisation, configured with an `ngram_range` of (1, 3) to capture unigrams to trigrams, allowing the model to represent short phrase-level patterns such as negation and local context [19]. The vocabulary was capped at 12,000 terms, with `min_df=2` and `max_df=0.9` used to exclude extremely rare and overly common terms, and standard English stopwords removal was applied to reduce the influence of high-frequency tokens.

TF-IDF was chosen because it weights terms by their importance within a document relative to the wider corpus [24]:

$$tf-idf(t, d) = tf(t, d) \cdot \log\left(\frac{N}{df(t)}\right)$$

where $tf(t, d)$ is frequency of term t in the document d , N is the total number of documents, and $df(t)$ is the number of documents containing term t . To prevent data leakage, the vectoriser was fitted exclusively on the training set and applied unchanged to the validation and test sets.

Prediction models

Three systems were developed and compared. All used Logistic Regression, configured with regularisation strength `C=2.0` and `max_iter=1000` to ensure convergence on the high-dimensional TF-IDF feature space [8]. Logistic Regression was selected because it is computationally efficient, interpretable, and provides a strong baseline for sparse high-dimensional text data [8].

- ❖ System 1 (Baseline) was trained on all 11,503 training samples without spam filtering, providing a reference point for comparison.
- ❖ System 2 (Clean-only) used the same architecture but was trained only on samples not flagged as likely spam, testing whether reducing label noise improved the decision boundary.
- ❖ System 3b (Final) extended System 2 with an explicit spam-handling stage at inference time. Samples flagged by the heuristic spam detector were assigned the dummy label -1 without being passed to the

classifier, while samples not flagged as spam were classified normally. This decoupled anomaly detection from sentiment classification and prevented arbitrary predictions on out-of-domain inputs.

Table 1: Summary of the *sentiment-classification systems* compared in Task 1

System	Training Data	Spam Handling	Key Purpose	Model
System 1 (Baseline)	All 11,503 samples	None	Standard supervised baseline	TF-IDF + Logistic Regression
System 2 (Clean-only)	Non-spam samples only	Likely spam removed during training	Reduce label noise	TF-IDF + Logistic Regression
System 3 (Spam-aware)	All samples	Dummy label (-1) at prediction time	Separate spam detection from sentiment prediction	TF-IDF + Logistic Regression + confidence/rules

Spam detection strategy

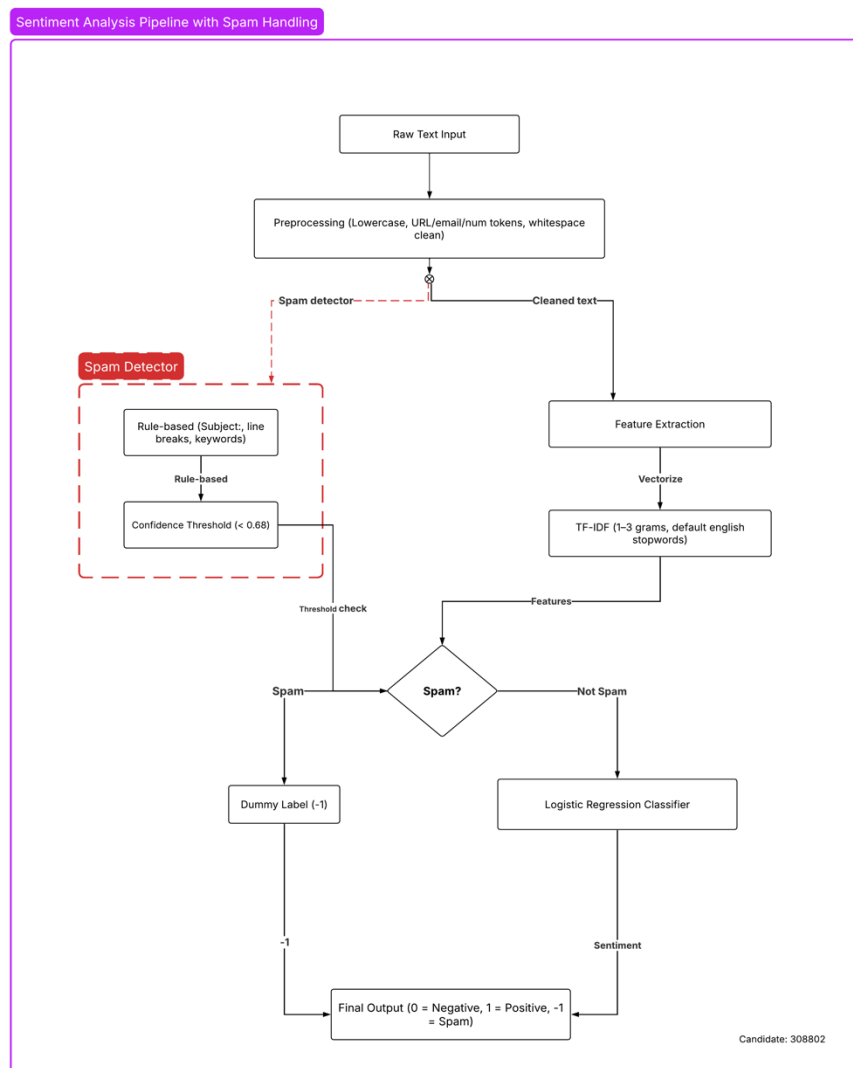
Likely spam instances were identified using deterministic heuristics derived from exploratory analysis of the raw text. A sample was flagged as likely spam if it satisfied any of the following heuristics applied to the raw text:

1. It began with the prefix “Subject:” (case-insensitive), which is strongly indicative of email-style headers.
2. It contained repeated email-style line breaks such as `\r\n` or `\n\n`.
3. It contained promotional keywords associated with spam-like content, such as “free”, “offer”, “click”, “buy”, “winner”, or “urgent”.

In addition to these deterministic rules, model confidence was used as a secondary signal. Any sample with a maximum predicted class probability

below 0.68 was also treated as likely spam, on the assumption that uncertain predictions often reflect out-of-domain or contaminated inputs [8][19]. Together, these criteria identified approximately 25.8% of the training set as likely spam. Flagged samples were excluded from training in System 2 and assigned the dummy label -1 at inference time in System 3.

Figure 2.1.2. *Overview of the spam-aware sentiment analysis pipeline*



Compute usage

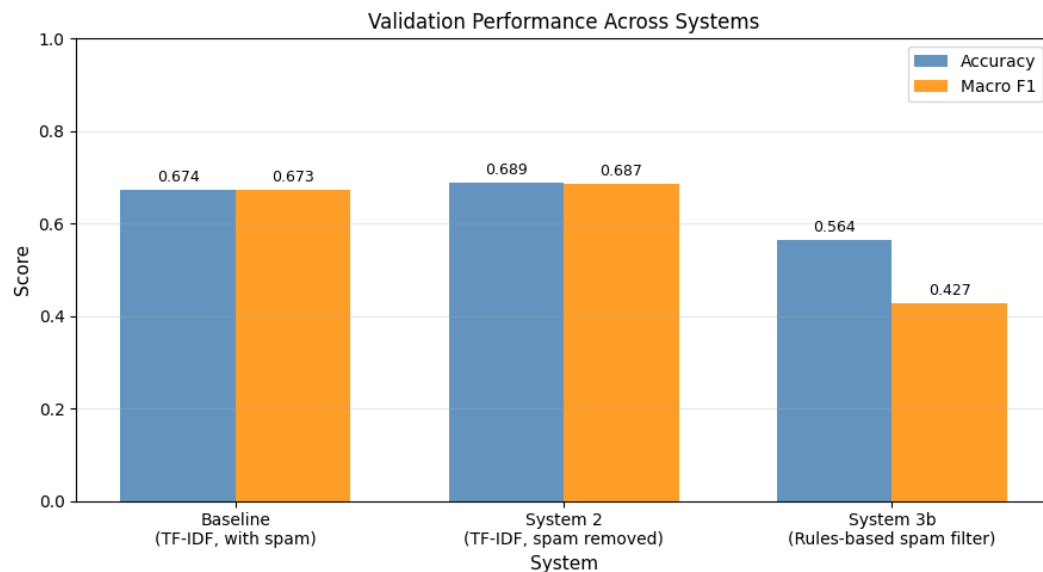
All experiments were conducted in Google Colab on CPU, as the models were computationally lightweight and did not require GPU acceleration. Training Logistic Regression on the full TF-IDF representation took approximately 7–12 seconds, while the complete spam-aware pipeline finished in under one minute, showing that the approach demonstrated computation efficiency despite the high-dimensional sparse feature space.

2.2 ANALYSIS OF RESULTS (HELD-OUT VALIDATION SET)

QUANTITATIVE EVALUATION

Task 1 was evaluated on the held-out validation set using the Baseline, System 2, and the final spam-aware System 3b. The Baseline achieved 0.6736 accuracy and 0.6734 macro F1, while System 2 improved to 0.6893 accuracy and 0.6867 macro F1 after removing likely spam from training, making it the strongest pure sentiment classifier on the full validation set. System 3b, however, was designed as the final submission model rather than the best ordinary validation classifier, so its use of dummy labels for likely spam reduced full-validation performance to 0.5641 accuracy and 0.4266 macro F1 when those predictions were counted as incorrect.

Figure 2.2.1 Validation set accuracy and macro F1-score across the three main systems.



When evaluated only on the clean validation subset, System 3b achieved 0.7392 accuracy and 0.7392 macro F1 while flagging 23.7% of samples as spam. This rate was close to the 25.8% likely-spam proportion identified during exploratory analysis, making System 3b more plausible as a final submission system than a standard classifier forced to assign only positive or negative labels to every sample.

An earlier hybrid version, System 3a, combined heuristic spam rules with a low-confidence threshold but was rejected because it was too aggressive. At a threshold of 0.68 it flagged 87.9% of the validation set as spam and achieved only 0.5030 accuracy and 0.4650 macro F1 on the remaining subset; even at 0.60 it still marked 68.1% as spam, despite raising clean-subset accuracy and macro F1 to 0.7303. This suggests that confidence-based spam rejection removed too many difficult but genuine reviews and produced a misleadingly narrow evaluation set.

Table 2.2.1: Validation performance comparison across the Baseline, System 2, System 3a, and System 3b, including full-validation and clean-validation metrics where applicable

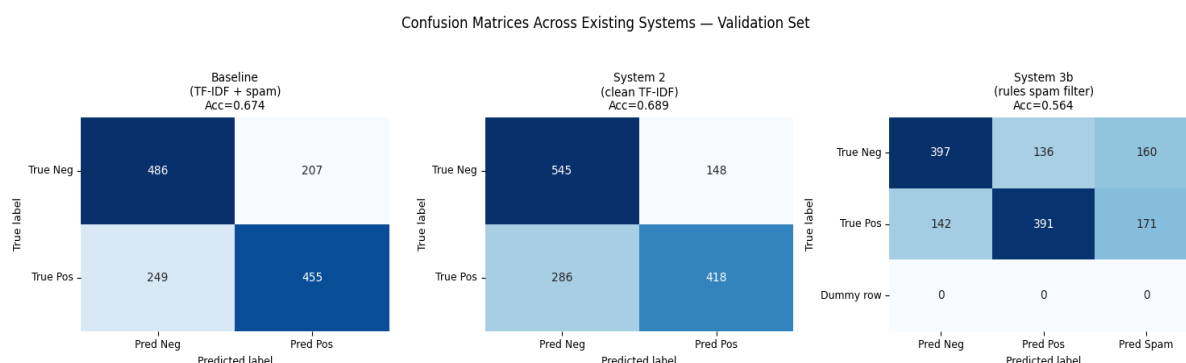
System	Training Data	Spam Handling Strategy	Full Validation Acc / Macro F1	Clean Validation Acc / Macro F1	Validation Samples Flagged as Spam
System 1 (Baseline)	All samples	No explicit spam handling	0.6736 / 0.6734	—	—
System 2 (Clean-only)	Likely non-spam only	Spam removed before training	0.6893 / 0.6867	—	—
System 3a (Hybrid)	All samples	Heuristic rules plus confidence threshold	—	0.5030 / 0.4650 at 0.68; 0.7303 / 0.7303 at 0.60	87.9%; 68.1%
System 3b (Final)	Likely non-spam only, using the	Heuristic rules only, with dummy	0.5641 / 0.4266	0.7392 / 0.7392	23.7%

System 2 sentiment model	spam label (-1)
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Qualitative Analysis and Failure Cases Analysis

The confusion matrices (Figure 2.2.2) reveal that the Baseline model produced slightly more false negatives than false positives. This pattern is consistent with spam-contaminated training data weakening the lexical association between positive sentiment and strongly affirmative vocabulary

Figure 2.2.2. Confusion matrices for the retained validation systems, showing the error patterns of the Baseline, System 2, and System 3b



System 2 reduced false positives substantially but increased false negatives, suggesting that cleaner training data produced a sharper boundary biased toward negative predictions while positive sentiment remained harder to classify reliably. In many cases, positive reviews were expressed through qualified, restrained, or mixed language rather than strongly positive vocabulary, making them less compatible with a sparse TF-IDF representation.

The misclassified validation examples suggest that the main weaknesses of the retained systems were linguistic rather than purely statistical. False negatives often involved mixed or concessive sentiment, where negative lexical cues appeared in reviews with an overall positive judgement, while false positives were often driven by negation, rhetorical reversal, or contrastive criticism. Table 2.2 shows that these issues persisted even in the final System 3b pipeline, indicating that rules-based spam filtering improved robustness to obvious spam but did not remove the core challenge of compositional sentiment interpretation.

Table 2.2.2. Example validation misclassifications and their likely causes.

Type	System	Example (truncated)	Likely cause
False negative	System 2	<i>“a very pretty after-school special. it’s an effort to watch this movie, but it eventually pays off and is effective if you stick with it.”</i>	Concessive structure; early negative cues may outweigh the overall positive judgement.
False negative	System 2	<i>“baby-faced renner is eerily convincing as this bland blank of a man with unimaginable demons within.”</i>	Positive judgement is expressed through words like “convincing,” but strongly negative lexical cues such as “bland” and “demons” may have pushed the model toward a negative prediction.
False positive	System 2	<i>“a guilty pleasure at best, and not worth seeing unless you want to laugh at it.”</i>	Contrastive criticism; mildly positive phrasing may be overweighted despite an overall negative judgement.
False positive	System 2	<i>“a sometimes incisive and sensitive portrait that is undercut by its awkward structure and a final veering toward melodrama.”</i>	Opening positive descriptors may dominate even though the final evaluation is negative.
Residual error (after spam filtering)	System 3b	<i>“there isn’t one moment in the film that surprises or delights.”</i>	Negation scope remained difficult to capture with TF-IDF n-gram features.

The text preprocessing aimed to preserve important sentiment cues such as negation; however, the TF-IDF vectoriser used the built-in English stop-word list, which likely removed tokens such as “not” and “no”. The use of 1–3-gram features partially compensate for this because phrases like “not good” or “no fun” can still appear as bigrams or trigrams, but this remains a limitation and likely contributes to some misclassifications in Table 2.2.

Overall, the final system performed best when sentiment was expressed directly through unambiguous vocabulary, and worst when interpretation depended on negation, concessive structure, sarcasm, or domain mismatch. System 2 was the strongest standard sentiment classifier, while System 3b was the most defensible final submission model for the assignment as defined.

2.3 EXTERNAL DATASET PERFORMANCE (NLTK MOVIE REVIEWS)

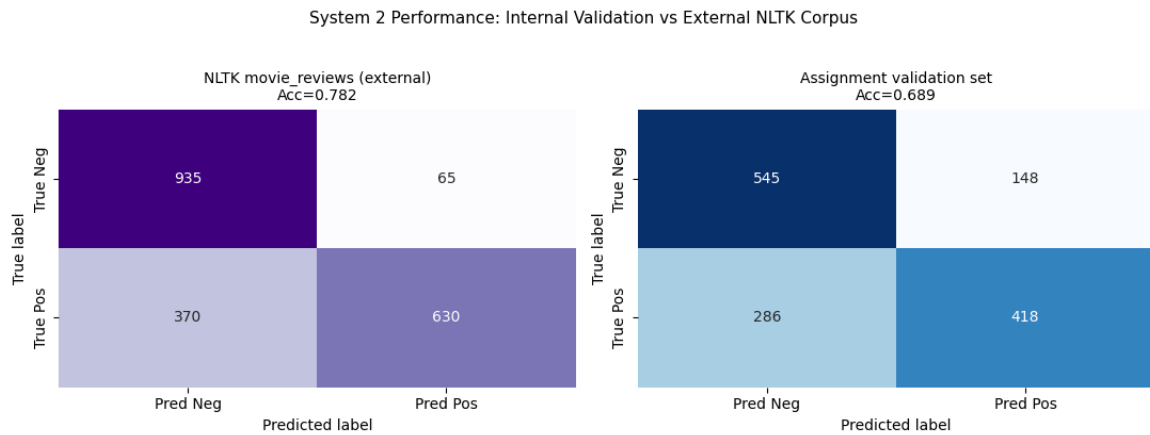
System 2 was also evaluated on the NLTK movie_reviews corpus to assess generalisation on clean external movie-review data. This dataset contains 2,000 balanced positive and negative reviews and does not include spam, making it a fairer test of pure sentiment classification than the assignment dataset. System 2 achieved 0.7825 accuracy and 0.7773 macro F1 on this external corpus, compared with 0.6893 accuracy on the assignment validation set.

Although System 3b was selected as the final model for the assignment test set because it explicitly handled spam, System 2 was more appropriate for external evaluation because the NLTK corpus contains only genuine sentiment examples. This allowed the generalisation test to focus on the quality of the sentiment classifier itself rather than on the usefulness of the spam-handling component.

Figure 5 compares the NLTK confusion matrix with the assignment validation confusion matrix for System 2. The model performed better on NLTK than on the assignment validation set, which is plausible because the NLTK corpus consists of well-formed movie reviews, whereas the assignment data included substantial spam-like material and noisy non-review text. The NLTK confusion matrix still showed asymmetric behaviour, with stronger recall for negative

reviews than for positive reviews, suggesting that even on clean external data the model was more reliable at recognising negative sentiment.

Figure 2.3.1. Confusion matrices for System 2 on the NLTK movie_reviews corpus and the assignment validation set, showing stronger performance on the cleaner external dataset.



Overall, this external evaluation suggests that the underlying sentiment model learned transferable sentiment cues rather than merely fitting assignment-specific noise. It also supports the view that the spam-handling component in System 3b was mainly necessary for the noisier assignment dataset

2.4 EXTENDED CRITICAL ANALYSIS AND LIMITATIONS

Critical Observations

1. The main finding was that data quality mattered more than model complexity, because spam-like content in the training data weakened the Baseline classifier's ability to learn genuine sentiment patterns.
2. System 2 showed that removing likely spam improved sentiment classification, while System 3a showed that overly aggressive filtering could remove genuinely ambiguous reviews rather than true spam.

3. System 3b was the most appropriate final submission system because it explicitly handled contaminated data, achieving 0.7392 accuracy on the clean validation subset while flagging 23.7% of validation samples as spam.
4. Although System 2 was the strongest pure sentiment classifier, System 3b was more suitable for the assignment because the hidden test set required spam-aware predictions.

Limitations and Future Work

1. The main strength of the final system was explicit spam handling, but its heuristic spam detector relied on simple rules and may miss novel spam or incorrectly flag genuine reviews.
2. The TF-IDF and Logistic Regression pipeline mainly captured surface-level lexical patterns, so it remained vulnerable to sarcasm, subtle negation, and mixed sentiment.
3. Future work could explore learned spam detection, a custom stop word function, richer text representations such as GloVe or transformer-based models, and deeper error analysis of difficult validation cases.

3 PART 2: FACE ALIGNMENT IN IMAGES

3.1 METHODOLOGY

Problem Formulation

Face alignment is formulated as a multi-output regression task. Given an input face image, the model predicts the (x, y) coordinates of 5 facial landmarks (left eye, right eye, nose, left mouth corner, right mouth corner), producing an output of shape (5, 2). The central challenge is the dataset's significant geometric and photometric distortions (rotations, scaling, perspective warps, and brightness shifts). A naive model trained without explicit robustness measures will fail under these variations. All design decisions below were made to address this variability.

Preprocessing Pipeline

The same preprocessing steps are applied to training, validation, and test sets (augmentation is applied during training only).

Figure 3.1.1. Preprocessing Pipeline

Preprocessing Pipeline

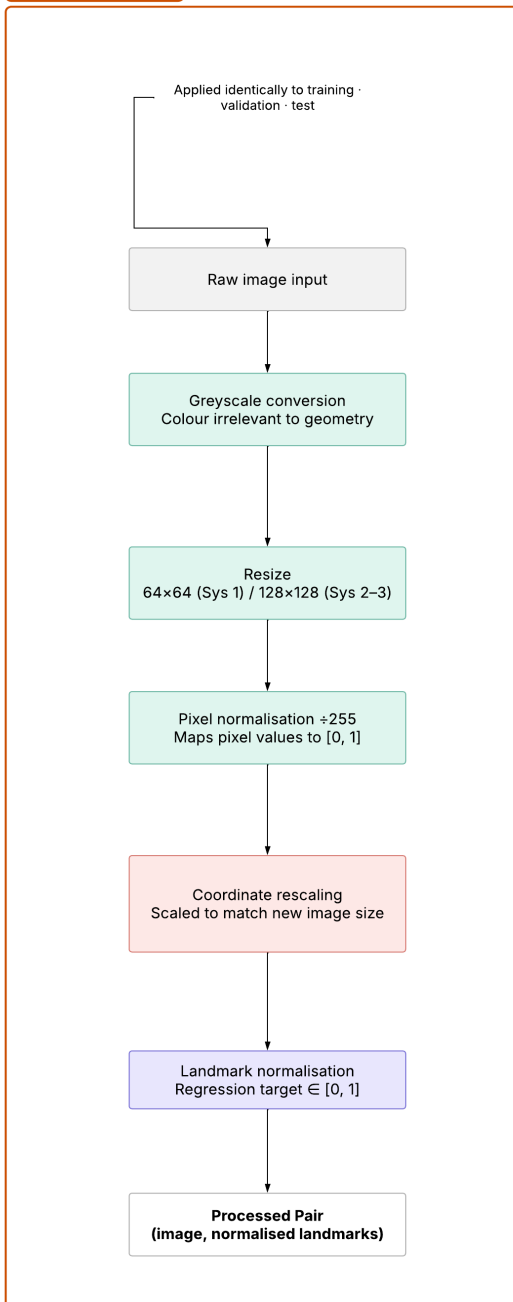


Figure 3.1. Interpretation

- ❖ **Grayscale conversion:** Facial landmark geometry is defined primarily by luminance and edges rather than colour, reducing input channels and computation by one-third with negligible performance loss.
- ❖ **Resize:** Reduces computation while preserving sufficient spatial resolution for accurate localisation, and ensures fair comparison between the HOG baseline and CNN.
- ❖ **Pixel normalisation ($\div 255$):** Maps values to $[0, 1]$ and prevents gradient saturation.
- ❖ **Landmark coordinate rescaling:** Coordinates were rescaled consistently with the resized images. For an image of width, W , and height, H , coordinates were normalised as:

$$x' = x \times \frac{W_{new}}{W_{original}}, y' = y \times \frac{H_{new}}{H_{original}},$$

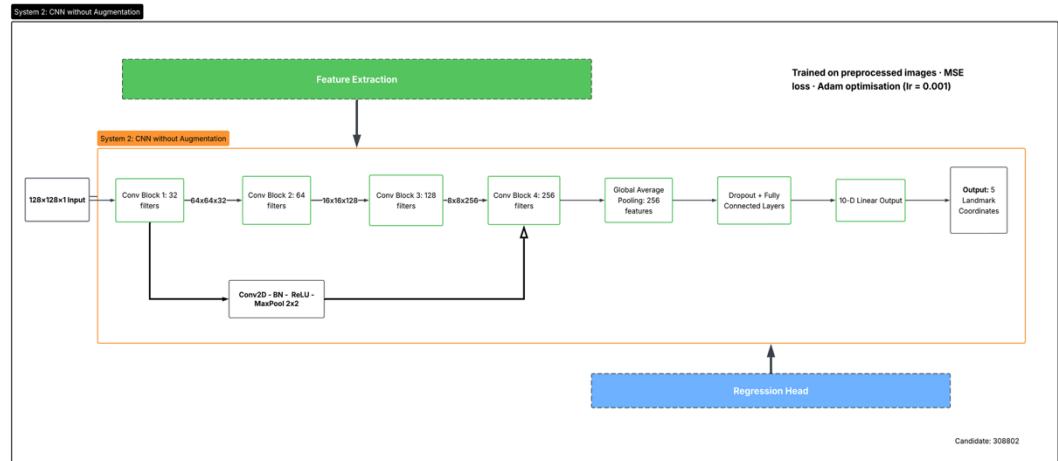
- ❖ **Coordinate normalisation to $[0, 1]$:** Produces a scale-invariant regression target and makes MSE loss values directly interpretable.

At inference time, predicted normalised coordinates were converted back to the original 256×256 pixel coordinate system before generating the results CSV.

System 1 — Baseline: HOG Features + Ridge Regression

HOG descriptors were extracted from 128×128 grayscale images using 9 orientations, 8×8 pixels per cell, and 2×2 cells per block, producing 8100-dimensional feature vectors that capture local edge structure and offer partial robustness to illumination changes [2]. These features are fed into a Ridge regressor ($\lambda = 1.0$, selected via cross-validation) to predict the 10 landmark coordinates simultaneously [6] [21]. This system provides a fast, interpretable baseline but is limited by its sensitivity to geometric distortions, motivating the shift to learned features.

Figure 3.1.2. System 2 — CNN Regressor A custom CNN learns hierarchical spatial features directly from preprocessed grayscale pixels [10] [20].



System 2 CNN Architecture

System 2 is a compact CNN that extracts hierarchical spatial features from 128×128 grayscale images [20], built around four convolutional blocks with progressively expanding channel sizes (32→64→128→256). Each block consists of a 3×3 convolution followed by Batch Normalisation, ReLU activation, and 2×2 MaxPooling. Rather than flattening the feature maps as in the earlier design, a Global Average Pooling head is used, which reduces the parameter count and improves spatial invariance. The resulting features are passed through Dropout ($p = 0.3$) and two fully connected layers, yielding 10 outputs that encode five (x, y) landmark coordinate pairs.

Training minimises MSE loss,

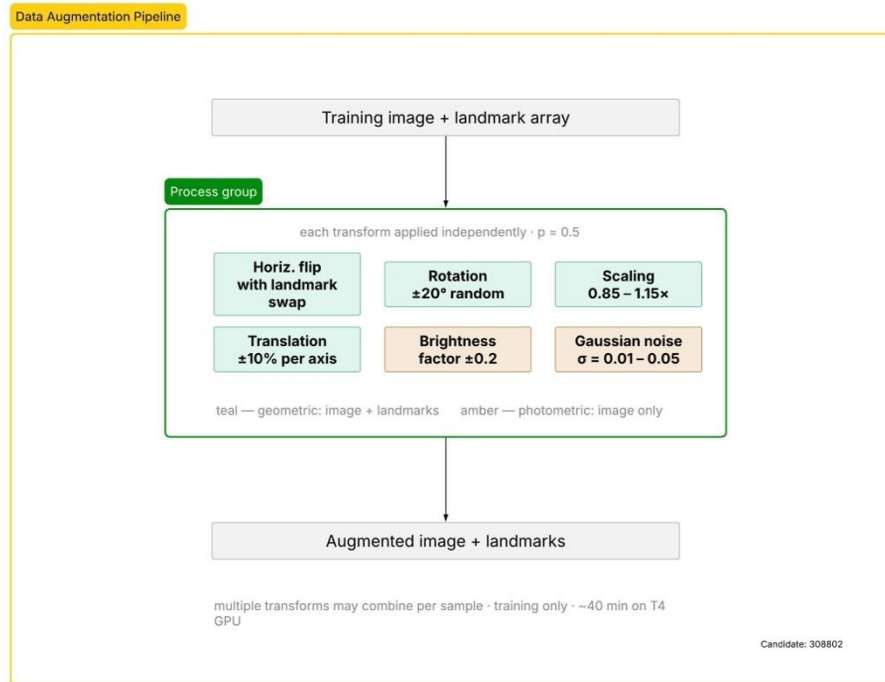
$$L = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^5 [(x_{ij} - \hat{x}_{ij})^2 + (y_{ij} - \hat{y}_{ij})^2]$$

where (x_{ij}, y_{ij}) and $(\hat{x}_{ij}, \hat{y}_{ij})$ denote the ground-truth and predicted coordinates of the j -th landmark in the i -th image. MSE penalises larger localization errors more strongly than L1 and converged faster in preliminary experiments. Optimisation uses Adam [7] (lr = 0.001, batch size = 64) with ReduceLROnPlateau scheduling, validation checkpointing and early stopping [11]. Compared with the Flatten-based CNN this compact architecture generalises better, has fewer parameters, and trains in about 1 minute on a T4 GPU when trained for a maximum of 40 epochs.

System 3 — Final Model: CNN with Data Augmentation

System 3 uses the same architecture as System 2 but incorporates a joint image-and-landmark augmentation pipeline during training to directly tackle the dataset’s geometric variability [13].

Figure 3.1.3: Data Augmentation + Regression Pipeline



Augmentations (applied with $p=0.5$): random horizontal flip (with corresponding coordinate and landmark swapping), rotation ($\pm 20^\circ$), scaling ($0.85\text{--}1.15\times$), translation ($\pm 10\%$), brightness jitter (± 0.2), and Gaussian noise ($\sigma=0.025$). Geometric transforms are applied consistently to both images and landmarks to preserve the supervision signal.

This augmentation strategy was designed to improve robustness to the distortions present in the test set. Training time increased to approximately 2 minutes per run (maximum 60 epochs) due to augmentation overhead.

System Comparison

Table 2.3. Summary of Developed Systems

System	Feature Representation	Model	Augmentation	Input Size	Approx. Parameters	Training Time (T4 GPU)	Key Strength	Key Limitation
System 1 (Baseline)	HOG (8100-dim)	Ridge Regression	None	64×64	$\sim 80\text{k}$	< 1 minute	Fast, interpretable	Sensitive to geometric distortion
System 2	Raw pixels	Custom CNN	None	128×128	$\sim 422\text{k}$	~ 1 minute	Learns hierarchical features	Limited robustness to distortions
System 3 (Final)	Raw pixels	Custom CNN	Full pipeline	128×128	$\sim 423\text{k}$	~ 2 minutes	Best accuracy + robustness	Higher computational cost

Evaluation

The primary quantitative metric used is Mean Euclidean Distance (MED) in normalised coordinates [0,1]. MED directly measures the average localisation error across the five predicted facial landmarks, making it well-suited for regression-based face alignment [22].

$$MED = \frac{1}{N} \frac{1}{5} \sum_{i=1}^N \sum_{j=1}^5 \sqrt{(x_{ij} - \hat{x}_{ij})^2 + (y_{ij} - \hat{y}_{ij})^2}$$

Robustness was assessed by applying controlled test-time perturbations to validation images — rotation, scaling, Gaussian noise, and brightness shifts at increasing severity levels — with the resulting change in MED used to quantify resilience to geometric and photometric variation. Qualitative evaluation uses overlaid landmark visualisations for best-case, median, and failure examples to surface patterns not captured by summary statistics alone.

Implementation

All convolutional models were implemented in PyTorch [23] and trained on a single NVIDIA T4 GPU (16 GB) in Google Colab, with NumPy-based preprocessing and scikit-learn used for the HOG + Ridge baseline [21].

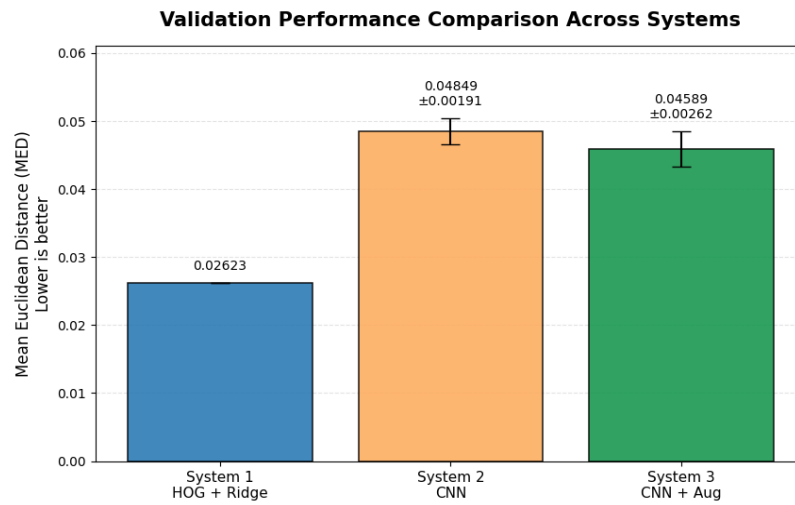
3.2 ANALYSIS OF RESULTS (HELD-OUT VALIDATION SET)

Quantitative Results

Lower MED indicates better landmark localisation. The HOG + Ridge baseline achieved the strongest overall performance, with a validation MED of 0.02623. The unaugmented CNN, System 2, achieved a mean validation MED of approximately 0.04849 across five runs, while the augmented CNN, System 3, achieved a lower mean validation MED of approximately 0.04589.

System	Validation MED	Notes
System 1: HOG + Ridge	0.02623	Best overall validation score
System 2: CNN	0.04849 0.00191	± Learned visual features but weaker than HOG
System 3: CNN + Augmentation	0.04589 0.00262	± Improved over System 2 but still below HOG

Figure 3.2.1. Validation performance comparison across the three developed systems.



System 1 achieved the lowest MED; System 3 improved over System 2 but did not surpass the HOG baseline.

Across the repeated runs, System 3 reduced MED by approximately 5.36% compared with System 2 on average. However, both CNN systems remained worse than the HOG baseline. System 2 was approximately 84.88% worse than HOG, while System 3 was approximately 74.97% worse than HOG. This suggests that, for this relatively small dataset, the hand-crafted HOG representation generalised more effectively than the CNN models.

Table 3.2.2. Repeated Run Comparison

Run	System2 MED	System3 MED	System 3 Change vs System 2
Run 1	0.04892	0.04147	+15.23% improvement
Run 2	0.04795	0.04732	+1.31% improvement
Run 3	0.04981	0.04802	+3.59% improvement
Run 4	0.04548	0.04564	-0.35% worse
Run 5	0.05031	0.04702	+6.54% improvement

System 3 improved over System 2 in four out of five runs. This indicates that data augmentation generally helped the CNN, although the improvement was not completely consistent across random initialisations and training runs.

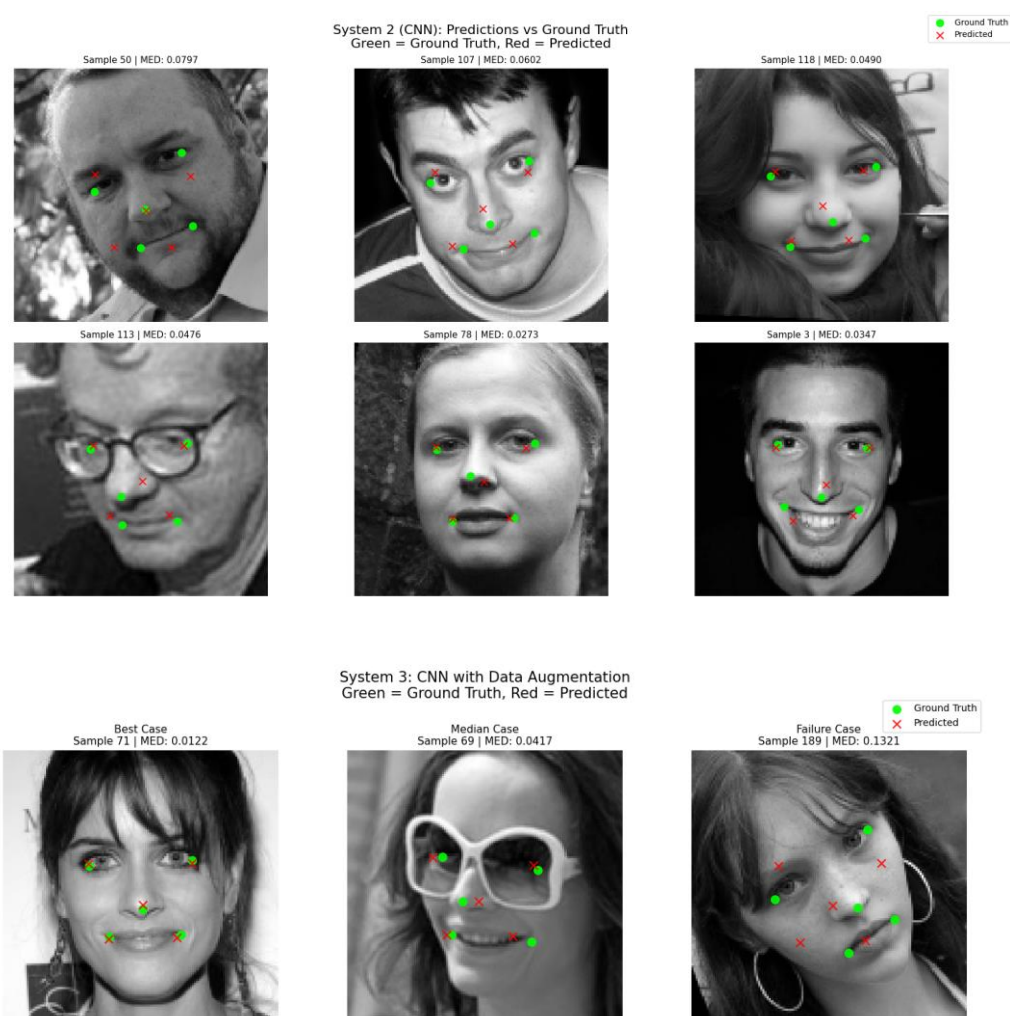
Per-Landmark Performance

The per-landmark MED values showed that the eyes were usually predicted more accurately than the nose and mouth corners. For example, in the fifth System 3 run, the left and right eye errors were around 0.03588 and 0.03871, while the nose error was higher at approximately 0.05943. This pattern suggests that eye landmarks were easier for the model to identify, probably because they have more distinctive local visual structure. The nose and mouth corners were more difficult, likely because they are more affected by pose, expression, lighting, and partial occlusion.

Qualitative Evaluation

System 3 performed well on clear, front-facing examples, while median cases showed small localisation errors, especially around the nose and mouth corners. Failure cases were mainly associated with strong rotation, poor illumination, partial occlusion, or severe geometric distortion. These visual results support the quantitative findings: augmentation improved the CNN compared with System 2, but difficult geometric cases remained challenging.

Figure 3.3.1. Run 5 Qualitative Predictions for System 2 and System 3 (Best/Median/Failure)



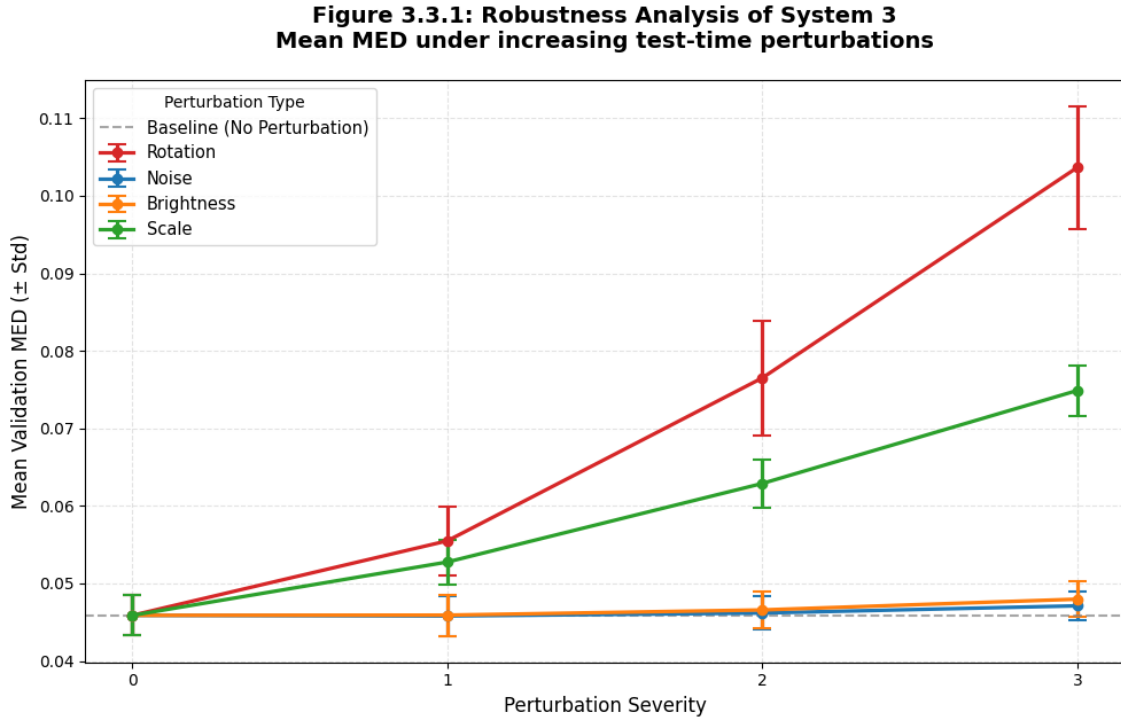
3.3 ROBUSTNESS ANALYSIS

Mean robustness was evaluated across five System 3 runs by applying controlled test-time perturbations to the validation set and measuring the average change in MED.

The average robustness pattern from System 3 was:

Perturbation	Severity 0	Severity 1	Severity 2	Severity 3	Main Trend
Rotation	0.04589 ± 0.00262	0.04589 ± 0.00262	0.07651 ± 0.00740	0.10365 ± 0.00790	Strong degradation
Noise	0.04589 ± 0.00262	0.04581 ± 0.00256	0.04619 ± 0.00215	0.04712 ± 0.00188	Small degradation
Brightness	0.04589 ± 0.00262	0.04581 ± 0.00256	0.04619 ± 0.00215	0.04798 ± 0.00232	Mild degradation
Scale	0.04589 ± 0.00262	0.05277 ± 0.00292	0.06290 ± 0.00314	0.07489 ± 0.00324	Clear degradation

Figure 3.3.1. Mean robustness analysis for System 3 across five runs.



The model was relatively robust to Gaussian noise and brightness changes, with severity-3 MED increasing only slightly from 0.04589 to 0.04712 and 0.04798 respectively. In contrast, geometric perturbations caused much larger degradation: rotation increased MED to 0.10365, while scaling increased it to 0.07489 at severity 3. This indicates that, although System 3 used geometric augmentation, strong rotation and scaling remained the main sources of localisation error.

3.4 EXTENDED CRITICAL ANALYSIS AND LIMITATIONS

The HOG + Ridge baseline proved most effective, indicating that hand-crafted gradient features generalised better than CNNs trained from scratch on this relatively small dataset (2,600 training images) and simplified 5-landmark task. Although data augmentation in System 3 provided a modest but generally consistent improvement over System 2, neither deep learning model surpassed the classical baseline.

Key limitations include the small dataset size, which restricted the CNNs' ability to outperform hand-crafted features, and moderate run-to-run variability (± 0.002 – 0.003 MED) were likely due to random weight initialisation and data shuffling. Moreover, the relatively simple architecture and direct coordinate regression approach may have limited the model's ability to capture spatial uncertainty. Geometric distortions, particularly rotation and scaling, remained the most challenging despite augmentation facial structure available for learning.

Future Work

Future work could explore mitigating the dataset size constraint through transfer learning from pretrained face models such as VGGFace [1], FaceNet[23], or ArcFace [3], which provide rich facial representations that require less task-specific data to fine-tune. Replacing direct coordinate regression with heatmap-based landmark prediction [26] would better capture spatial uncertainty and has been shown to improve localisation accuracy on challenging geometric distortions.

To address the rotation and scaling sensitivity specifically, more targeted augmentation strategies such as TPS (Thin Plate Spline) warping or perspective transforms could supplement the current pipeline. Training on larger benchmarks such as 300-W, WFLW, or AFLW would allow deeper architectures to reach their full potential and provide a more rigorous evaluation framework. Finally, ensemble methods averaging predictions across multiple independently trained models would reduce the run-to-run variability.

Overall, this project demonstrates the value of strong baselines and critical evaluation — sometimes the simplest approach remains the most effective.

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